

Huaiyuan Jing

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EDUCATION

University of Illinois Urbana-Champaign <i>Master of Computer Science</i>	Urbana-Champaign, IL <i>Aug 2026 — Present</i>
University of Wisconsin-Madison <i>B.S. Computer Science; B.S. Mathematics</i>	Madison, WI <i>Sep 2022 — May 2026</i>
- GPA: 3.8/4.0 US Employment Authorization: Yes - Relevant Coursework: Algorithms, Operating Systems, Quantum Architecture, Linear Algebra, Probability Theory	

HONORS & AWARDS

Gold Medal, ICPC North Central North America Regional (Top 1%)	2024–2025
Silver Medal, ICPC North Central North America Regional	2025–2026
Ranked elite among university teams, excelling in dynamic programming, advanced data structures, and edge-case handling under extreme time constraints and psychological pressure.	

EXPERIENCE & RESEARCH

Research Assistant Madison Experimental Mathematics (MXM) Lab	Feb 2025 — Present <i>Madison, WI</i>
- Architected a high-throughput, concurrent toolkit to compute cryptographic hash values for complex graphs and evaluate combinatorial game strategies (Grundy numbers), successfully evaluating 50 critical data points under extreme $O(2^m)$ time-complexity workloads. - Leveraged Rust’s Send/Sync traits and ownership model to eliminate data races and deadlocks, designing memory-safe parallel processing streams that maximize multi-core CPU and GPU saturation. - Optimized graph traversal routines with low-latency lock-free structures, minimizing algorithmic latency under high-complexity, multi-threaded workloads.	
First Author Quantum Computing Algorithm Optimization Research	Summer 2026 <i>Madison, WI</i>
- Proposed an optimized method for single-ancilla block-encoding overhead reduction, successfully reducing the required auxiliary qubit overhead to exactly 1 ancilla qubit via polar decomposition. - Coherently reused block-encoding ancilla to produce a unified $(1, 1, \epsilon)$-block encoding for general matrices, cleanly distributing a total error budget of ϵ into two $\frac{\epsilon}{2}$ branches to guarantee strict linear error propagation. - Conducted rigorous query complexity and mathematical error bounds analysis, bounding the polynomial degree for absolute-value approximation to $O\left(\frac{1}{\delta_{\text{abs}}} \log\left(\frac{1}{\eta_{\text{abs}}}\right)\right)$ to prove optimal computational scaling.	

PROJECTS

Independent Developer , High-Frequency Trading (HFT) Matching Engine & Order Book (codeberg.org/HYJING/hft_orderbook)	Jul 2026
- Designed and implemented an ultra-low latency, fixed-point limit order book (LOB) matching engine from scratch using Modern C++ (C++20). - Achieved an execution throughput of 38.55 million orders per second (38.5496M/s) under intensive multivariable cross-order stress tests, benchmarked via Google Benchmark on an 8-core CPU system. - Optimized memory layout by combining <code>std::map</code> for price levels with custom intrusive doubly-linked lists for strict FIFO order queues, minimizing lookup and allocation overhead to $O(1)$ for cancellation and insertion. - Engineered a vector-backed object-pool memory allocator (<code>VectorOrderBook</code>) to completely eliminate runtime OS heap allocations, preventing memory fragmentation and maximizing CPU L1/L2 data cache hit rates.	
Independent Developer , Linux Infrastructure & Kernel Optimization	Systems Automation
Maintained a security-critical initramfs module (Dracut-Cryptsetup-Duress) for LUKS-encrypted subsystems, demonstrating deep understanding of the Linux boot process and robust scripting.	

MOTIVATION LETTER

Dear Akuna Capital Recruitment Team,

I am writing to express my strong interest in participating in the **2026 Virtual Quant Trading Challenge**. As an incoming Master of Computer Science student at UIUC and a recent graduate with double Bachelor's degrees in Computer Science and Mathematics from UW-Madison (GPA 3.8/4.0), my primary motivation for joining this challenge stems from a deep fascination with Akuna Capital's core business model and the unique, adversarial nature of algorithmic market-making.

Unlike traditional quantitative funds that focus on directional alpha, an options market-maker thrives on the complex management of multi-dimensional risk and continuous liquidity provision. My interest in Akuna is driven by this exact challenge: pricing risk accurately in volatile environments while maintaining competitive, two-sided quotes. In options trading, success relies heavily on mathematical intuition—understanding how implied volatility shifts and how Greeks propagate through a portfolio. I am highly motivated to participate because I want to see how theoretical mathematical structures hold up when exposed to the relentless speed and toxic flow of a live simulated order book.

The Virtual Quant Trading Challenge presents the perfect arena to test my understanding of market microstructure. Designing a Python-based market-making bot is not merely a coding exercise; it is a dynamic game of inventory management and risk mitigation. In a simulated exchange competing against Akuna's models, the real test will be avoiding adverse selection and managing inventory skew. When asymmetric information hits the market, the bot must rapidly widen or skew its spreads to avoid being run over by informed traders. Having independently built a high-frequency matching engine and trained extensively to handle extreme psychological pressure during algorithmic programming regional championships, I possess a unique appreciation for order book dynamics and rapid decision-making under stress.

I view this challenge as the ultimate playground to bridge my mathematical foundations with execution strategy. I am fully prepared to dedicate 5 to 8 uninterrupted hours between August 17th and August 21st, 2026, to iterate, backtest, and optimize my trading bot. I look forward to competing against top talents and demonstrating my potential to contribute to Akuna's quantitative trading desks in Chicago. Thank you for your time and consideration.

Sincerely,

Huaiyuan Jing